

Assumption

Main standard ecommerce schema assume

kar raha hoon: orders, order_items,
products, customers. Isme orders.order_id,
orders.customer_id, orders.order_date,
orders.city, orders.region,
order_items.order_id,
order_items.product_id,
order_items.quantity,
order_items.unit_price,
products.product_id,
products.product_name use hoga
amazonaws .

1) Total Revenue

sql



```
SELECT
    SUM(oi.quantity * oi.unit_price)
AS total_revenue
FROM order_items oi;
```

2) Best-Selling Products

sql



```
SELECT
    p.product_id,
    p.product_name,
    SUM(oi.quantity) AS
total_units_sold,
    SUM(oi.quantity * oi.unit_price)
AS revenue
FROM order_items oi
JOIN products p
    ON oi.product_id = p.product_id
GROUP BY p.product_id,
p.product_name
ORDER BY total_units_sold DESC,
revenue DESC;
```

3) Revenue by City/Region

sql



```
SELECT
    o.city,
    o.region,
    SUM(oi.quantity * oi.unit_price)
AS revenue
FROM orders o
JOIN order_items oi
    ON o.order_id = oi.order_id
GROUP BY o.city, o.region
ORDER BY revenue DESC;
```

4) Peak Purchase Times

sql



```
SELECT
    EXTRACT(HOUR FROM o.order_date)
AS purchase_hour,
    COUNT(*) AS total_orders
FROM orders o
GROUP BY EXTRACT(HOUR FROM
o.order_date)
ORDER BY total_orders DESC,
purchase_hour;
```



```
-- KPI Queries
```

```
-- 1. Total Revenue
```

```
SELECT
```

```
    SUM(oi.quantity * oi.unit_price)
```

```
AS total_revenue
```

```
FROM order_items oi;
```

```
-- 2. Best-Selling Products
```

```
SELECT
```

```
    p.product_id,
```

```
    p.product_name,
```

```
    SUM(oi.quantity) AS
```

```
total_units_sold,
```

```
    SUM(oi.quantity * oi.unit_price)
```

```
AS revenue
```

```
FROM order_items oi
```

```
JOIN products p
```

```
    ON oi.product_id = p.product_id
```

```
GROUP BY p.product_id,
```

```
p.product_name
```

```
ORDER BY total_units_sold DESC,
```

```
revenue DESC;
```

-- 3. Revenue by City/Region

```
SELECT
    o.city,
    o.region,
    SUM(oi.quantity * oi.unit_price)
AS revenue
FROM orders o
JOIN order_items oi
    ON o.order_id = oi.order_id
GROUP BY o.city, o.region
ORDER BY revenue DESC;
```

-- 4. Peak Purchase Times

```
SELECT
    EXTRACT(HOUR FROM o.order_date)
AS purchase_hour,
    COUNT(*) AS total_orders
FROM orders o
GROUP BY EXTRACT(HOUR FROM
o.order_date)
ORDER BY total_orders DESC,
purchase_hour;
```

Indexing for Efficiency

Large datasets ke liye ye indexes useful rahenge:

sql



```
CREATE INDEX idx_orders_order_date
ON orders(order_date);
CREATE INDEX idx_orders_city_region
ON orders(city, region);
CREATE INDEX
idx_order_items_order_id ON
order_items(order_id);
CREATE INDEX
idx_order_items_product_id ON
order_items(product_id);
CREATE INDEX idx_products_product_id
ON products(product_id);
```


Optimization Notes

- `orders.order_id`, `order_items.order_id`, aur `order_items.product_id` par indexing join ko fast banati hai.
- `GROUP BY` aggregation ke liye query ko simple aur efficient rakha gaya hai.
- Agar data bahut bada ho, to `date` column par partitioning bhi helpful ho sakti hai.
- `order_date` se hour extract karna peak purchase time nikalne ke liye sufficient hai, but exact timezone consistency zaroor check karni chahiye `amazonaws`.

Sample Validation Outputs

Neeche example outputs hain jo sample data se match karne ke liye use kiye ja sakte hain:

Sample Validation Outputs

Neeche example outputs hain jo sample data se match karne ke liye use kiye ja sakte hain:

KPI	Sample Output
Total Revenue	125000.00
Best-Selling Product	Product A
Top City/Region	Mumbai / West
Peak Purchase Hour	20

Validation Query Example

Agar sample data test karna ho, to aap aise check kar sakte ho:

sql



```
SELECT
    COUNT(*) AS total_order_items,
    SUM(quantity) AS total_units,
    SUM(quantity * unit_price) AS
total_revenue
FROM order_items;
```

Agar aap chaho, main next message me isko **proper interview assignment format** me ek clean final document style me bana dunga, ya **single combined SQL file** ke roop me ready-to-copy version de dunga.